The documentation and process conversion measures necessary to comply with this amendment shall be completed by 17 June 2000.

INCH-POUND

MIL-PRF-19500/159K
AMENDMENT 1
17 March, 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE
TYPES 1N821-1, 1N823-1, 1N825-1, 1N827-1, AND 1N829-1, 1N821UR-1, 1N823UR-1, 1N825UR-1, 1N827UR-1, AND 1N829UR-1, JAN, JANTX, JANTXV, JANJ, JANS, JANHC AND JANKC; RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H; JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; AND JANKCM, D, L, R, F, G, H

This amendment forms a part of MIL-PRF-19500/159K, dated 19 November 1999, and is approved for use by all Departments and Agencies of the Department of Defense.

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Figure 4 Design Data, delete in it's entirety and replace with

"DESIGN DATA

Metallization:

Circuit layout data:
For zener operation, cathode must be operated positive with respect to anode.

Top: (Anode) . . . Al
Back . . . . . Au

Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ± 0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 4. JANHC and JANKC (B-version) die dimensions."

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Figure 5 Design Data, delete in it's entirety and replace with

"Metallization:

Circuit layout data:	Top:	2(Anode)	Al
For zener operation, cathode must be		1(Cathode)	Al
operated positive with respect anode.	Back	. . . . .	Au

Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Backside must be electrically isolated.

FIGURE 5. JANHC and JANKC (C-version) die dimensions."

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4.4.3.1. Group C inspection, appendix E, table VII of MIL-PRF-19500. Delete C2 in it's entirety

"C2 1056 Lead fatigue: Test condition E; tension: test condition A; 4 pounds weight, t = 15 ±3 seconds
2036 (lead fatigue and tension test are not applicable to surface mount "UR" version device).
1071 Condition E
1021 Omit initial conditioning "
and replace with,

"C2 1056
2036 Lead tension: Test condition Test condition A; 4 pounds weight, t = 15 ±3 seconds
Lead fatigue: Test condition E (lead tension and fatigue tests are not applicable to surface mount "UR" version device).
1071 Condition E
1021 Omit initial conditioning "

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Table II, Group D Inspection, subgroup 2, Voltage stability, delete method "4922" and replace with "4022".

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6.4., Delete "QPL" and replace with "QML".

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2270-03)

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